

Diabetic Retinopathy Telemedicine Outcomes With Artificial Intelligence-Based Image Analysis, Reflex Dilation, and Image Overread



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- **PURPOSE:** To examine real-world telemedicine outcomes of diabetic retinopathy (DR) screening with artificial intelligence (AI)-based image analysis, reflex dilation, and secondary image overread in a primary care setting.
- **DESIGN:** Validity and reliability analysis.
- **METHODS:** Single institution review of 1052 consecutive adult patients who received diabetic retinopathy photostereoscopy in the primary care setting over an 18-month period. Nonmydriatic fundus photographs were acquired and analyzed by the IDx-DR AI-based system. When nonmydriatic images were ungradable, reflex dilation (1% tropicamide) and mydriatic photography were performed for repeat AI-based analysis. Manual overread was performed on all images. Patient demographics, clinical characteristics, and screening outcomes were recorded.
- **RESULTS:** A total of 965 of 1052 patients (91.7%) had AI-gradable fundus photographs: 580 had gradable nonmydriatic imaging (55.1%) and 440 of 472 patients with ungradable nonmydriatic photographs had reflex dilation (93.2%). One hundred thirty-eight of 965 patients (14.3%) were AI-graded as “positive” (greater than mild NPDR) and 827 of 965 were “negative” (85.7%), with 100% sensitivity (95% CI 90.8-100%), 89.2% specificity (95% CI 87.0-91.1%), 27.5% positive predictive value (95% CI 24.0-31.4%), and 100% negative predictive value (95% CI 99.6-100%) compared with manual overread assessment of greater than mild NPDR requiring further evaluation with a comprehensive dilated examination. Image gradeability was inversely related to patient age: 93.5% gradable (61.9% nonmydriatic) for patients aged <70 years vs 85.3% (31.0% nonmydriatic) for patients aged 70+ years ($P < .001$).
- **CONCLUSION:** Incorporation of AI-based image analysis into real-world primary care diabetic retinopathy screening yielded no false negative results and offered excellent image gradeability within a protocol combining nonmydriatic fundus photography and pharmacologic

dilation, as needed. Image gradeability was lower with increasing patient age. (Am J Ophthalmol 2022;244: 125–132. © 2022 Elsevier Inc. All rights reserved.)

INTRODUCTION

DIABETIC RETINOPATHY (DR) IS A LEADING CAUSE OF avoidable visual loss, currently estimated to affect approximately 103 million people worldwide;¹ 2.9 million people are estimated to have moderate to severe visual impairment due to DR, while an additional 0.86 million are estimated to be functionally blind.² Compared with other causes of vision loss, DR disproportionately impacts the younger population and those living in higher income countries.^{3,4} Diabetic retinopathy is the leading cause of new legal blindness among adults aged 20 to 74 years in the USA.⁵ The prevalence of DR among Americans with diabetes mellitus (DM) is approximately 29% overall, but with even higher rates among non-Hispanic black patients and those with type 1 DM.^{4,7} Although systemic diabetes management has substantially improved in recent years, the global burden of DR-related vision loss continues to rise as a result of the increasing prevalence of diabetes worldwide.^{3,8-11}

Early detection and monitoring of DR is critically important, as the majority of DR-related blindness is preventable with timely management, including risk factor modification, intravitreal pharmacotherapy, laser photocoagulation, and surgery.¹¹⁻¹³ Regular screening for DR with dilated retinal examination and/or retinal imaging is recommended by the International Council of Ophthalmology, the American Academy of Ophthalmology (AAO), and the American Diabetes Association (ADA).^{3,12,14} Current AAO guidelines recommend DR screening at least annually starting 5 years after onset of type 1 diabetes, and at the time of diagnosis for patients with type 2 diabetes.¹⁴ Current ADA recommendations for initial DR screening are similar, with follow-up recommendations based on the presence and severity of retinopathy.¹⁵ However, the AAO estimates that 60% of patients with diabetes receive screening ex-

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aminations at least yearly.¹⁴ Recent studies have suggested that patients with no retinopathy and excellent glycemic control may have a relatively low risk of developing sight-threatening retinopathy at longer screening intervals; further investigation is warranted prior to considering implementing more complex and/or less frequent screening protocols.¹¹

An optimized DR screening system must offer consistent access to either dilated retinal examination or retinal imaging plus interpretation, minimized patient barriers for adherence to screening (eg, financial burden, inconvenient location, timing, or duration), and excellent communication of results and recommendations.^{11,12,16} Recent advances in retinal imaging and telemedicine technologies offer significant promise for improving screening adherence through expansion of potential screening locations and times that may be more convenient for patients, such as during primary care visits.^{3,11,12} Telemedicine is increasingly being incorporated into DR screening programs, but access to image interpretation remains limited in some systems.¹⁷ Artificial intelligence (AI)-based image analysis offers an opportunity for real-time, accurate, and scalable image interpretation.^{3,11,18,19} Real-time AI analysis-based feedback on image gradeability also offers an opportunity to reserve pharmacologic mydriasis for only those individuals who need it. This reduces a potential screening barrier for patients for whom dilation is unnecessary, while maximizing image quality and reducing unnecessary future visits for patients who do require dilation for safe DR surveillance.

The IDx-DR autonomous AI diagnostic platform (Digital Diagnostics) was designed to detect greater than mild DR through the analysis of 2 fundus photographs of each eye using the TRC 400NW nonmydriatic fundus camera (Topcon Corporation), and is the first such AI-based system approved for DR screening by the US Food and Drug Administration.¹⁹ The current study reports real-world outcomes following incorporation of the IDx-DR AI-based system into a pre-existing primary care-based DR telemedicine screening program with implementation of a reflex dilation protocol based on point-of-care image analysis results. Patient factors impacting image gradeability were analyzed, and manual image overread was performed on all photographs within the structure of the pre-existing Mayo Clinic Diabetic Retinopathy Telemedicine program.

METHODS

A single-institution retrospective review was performed of the electronic medical records of patients who received DR screening in the primary care setting over an 18-month period between September 2019 and March 2021. Research authorization was obtained from all patients included in the study. This retrospective study was approved by the Mayo

Clinic Institutional Review Board and performed in accordance with Health Insurance Portability and Accountability Act regulations. Institutional review board approval was obtained prior to initiation of the study. Study subjects included adult patients aged at least 18 years with DM, who received care within Mayo Clinic primary care clinics, and who received DR telemedicine screening within the Mayo Clinic Diabetic Retinopathy Telemedicine Program. For patients who had repeat screening during the study period, only the initial screening was included for analysis. Demographic information was recorded, including: patient age, sex, self-reported race/ethnicity, type of DM, and examination results. An “other” race/ethnicity category included patients who selected “other”, multiple race categories, or a category comprising < 1% of the study population. Electronic health records were reviewed for clinical characteristics including lens status, cataract severity, and presence of other ocular media opacities.

The Mayo Clinic Diabetic Retinopathy Telemedicine Program stipulates that all patients must have a comprehensive dilated eye examination within 3 years prior to initial telemedicine photostereotyping, and that telemedicine screening should not be utilized for patients with pre-existing moderate or severe DR, or for those with other known chronic ocular diseases including age-related macular degeneration, glaucoma, or uveitis. Primary care clinic laboratory technicians perform nonmydriatic color fundus photography with the Topcon TRC 400NW camera and submit two 45° 5-megapixel color fundus photographs from each eye centered on the optic nerve and macula for analysis. The IDx-DR AI-based system evaluates for the presence of greater than mild DR. If image quality is insufficient for analysis by the IDx-DR system and the technician assesses that improved nonmydriatic images are not readily achievable, a reflex dilation protocol is initiated and one drop of 1% tropicamide is administered to both eyes. Once the patient’s pupils are dilated, fundus photography is repeated and mydriatic photographs are resubmitted for IDx-DR grading.

Original fundus photographs (IDx-analyzed images plus any additional acquired photographs) and IDx-DR screening results were securely uploaded to the electronic health record system. Fundus photographs of patients with either positive or ungradable IDx-DR screening results were reviewed by an ophthalmologist or optometrist within 2 to 3 days. Fundus photographs of patients with “negative” screening results were compiled and batch-reviewed monthly. Reviewers had a minimum of 2 years of DR telemedicine image interpretation experience prior to the study period, and equivocal images were reviewed with a retina specialist (AJB) for adjudication. If significant DR or other retinal pathology requiring evaluation was identified, or if image quality was insufficient for screening purposes, the patient was referred for comprehensive dilated eye examination within an appropriate time frame. If no DR or only mild nonproliferative DR was present that did not

TABLE 1. Image Gradeability by Patient Sex, Race/Ethnicity, and Diabetes Mellitus Type

	<i>n</i> (%)	Gradable (%) Nonmydriatic	<i>P</i> Value	Gradable (%) Dilated	<i>P</i> Value	Ungradable (%) Declined Dilation
Overall	1052	580 (55)	—	965 (92)	—	32 (3)
Sex						
Male	631 (60)	329 (52)	<i>ref</i>	584 (93)	<i>ref</i>	17 (3)
Female	421 (40)	251 (60)	.02	381 (91)	.49	15 (4)
Race/Ethnicity						
White	815 (78)	463 (57)	<i>ref</i>	761 (93)	<i>ref</i>	21 (3)
Hispanic/Latin	18 (2)	10 (56)	.91	18 (100)	.62	0 (0)
Black	72 (7)	38 (53)	.50	63 (88)	.09	2 (3)
Other	118 (11)	54 (46)	.02	99 (84)	< .001	7 (6)
—	29 (2)					
Diabetes Mellitus Type						
Type 1	55 (5)	42 (76)	<i>ref</i>	50 (91)	<i>ref</i>	3 (5)
Type 2	994 (95)	533 (54)	.001	908 (92)	.50	29 (3)

Dilated = patients with gradable images after pupillary dilation protocol; Nonmydriatic = patients with gradable images prior to pupillary dilation protocol.

meet the clinical threshold to recommend a comprehensive dilated eye exam, repeat screening was recommended in 1 year. In accordance with the previously established Mayo Clinic DR telemedicine screening protocol, a repeat comprehensive dilated fundus exam was recommended for all patients at least every 3 years.

Outcome measures included the percentage of patients with AI-gradable images prior to and following reflex pupillary dilation, patient characteristics impacting image gradeability, as well as the sensitivity, specificity, positive predictive value (PPV), and negative predictive value of the IDx image interpretation in comparison with manual image review.

- **STATISTICAL ANALYSIS:** Categorical variables were expressed as percentages. Comparisons of these factors between groups was completed with a Chi-square test for independence. In addition, 95% confidence intervals were constructed for key estimates. *P*-values < .05 were considered significant. All analyses were completed with SAS version 9.4 (SAS Institute Inc.).

RESULTS

A total of 1204 consecutive unique patients received DR screening utilizing the IDx-DR AI-based platform during the study period, of whom 1052 had prior research authorization and were included in this study; 421 study subjects were female (40%) and 994 had type 2 DM (95%) (Table 1). Image quality was sufficient for IDx-DR AI-based image analysis in 965 of 1052 patients overall (91.7%), and 580 of 1052 had sufficient nonmydriatic image quality for analysis (55.1%). Among those with ungradable nonmydriatic photos, 440 of 472 proceeded to have reflex dilation

with 1% tropicamide (93.2%), and 385 of 440 had subsequent mydriatic fundus imaging that was gradable by the IDx-DR platform (87.5%).

Eight hundred twenty-seven of 965 patients with gradable retinal photographs (85.7%) had “negative” DR screening results with mild NPDR or less, and there was 100% manual overread agreement on these “negative” results (95% CI 99.6-100%). Of the 138 of 965 whose images were graded “positive” for more than mild NPDR (14.3%), manual overread identified 33 of 138 with at least moderate NPDR requiring further evaluation (23.9%), 3 of 138 with limited image quality for assessment (2.1%), and 2 of 138 with other retinal pathology (1.4%); all were referred for comprehensive dilated eye examination. One hundred of 138 of those with positive-graded images (72.4%) were assessed by manual overread to have relatively mild NPDR that was amenable to continued annual screening (Figure 1). Compared with manual review of fundus photographs, the AI-grading system had a sensitivity of 100.0% (95% CI 90.8-100.00%) and a specificity of 89.2% (95% CI 87.0-91.1%) for moderate or greater DR requiring further evaluation. The positive predictive value of AI grading was 27.5% (95% CI 24.0-31.4%) and negative predictive value was 100% (95% CI 99.6-100%).

Image gradeability was inversely related to patient age (Figure 2). Of all patients aged ≤ 70 years, 767/820 (93.5%) had IDx-DR gradable images (508/820 [62.0%] gradable nonmydriatic) compared with 198/232 (85.3%) of those aged 71+ with gradable images (72/232 [31%] nonmydriatic, *P* < .001 for overall and nonmydriatic comparisons). Nonmydriatic image gradeability varied significantly by age cohort, and was gradable for 67 of 81 patients aged 18 to 40 years (82.7%) compared with 82 of 114 patients aged 41 to 50 years (71.9%; *P* = .08), 168 of 246 patients aged 51 to 60 years (68.3%; *P* = .01), 191 of 379 patients aged 61 to 70 years (50.4%; *P* < .001), 65 of 196 patients aged 71 to

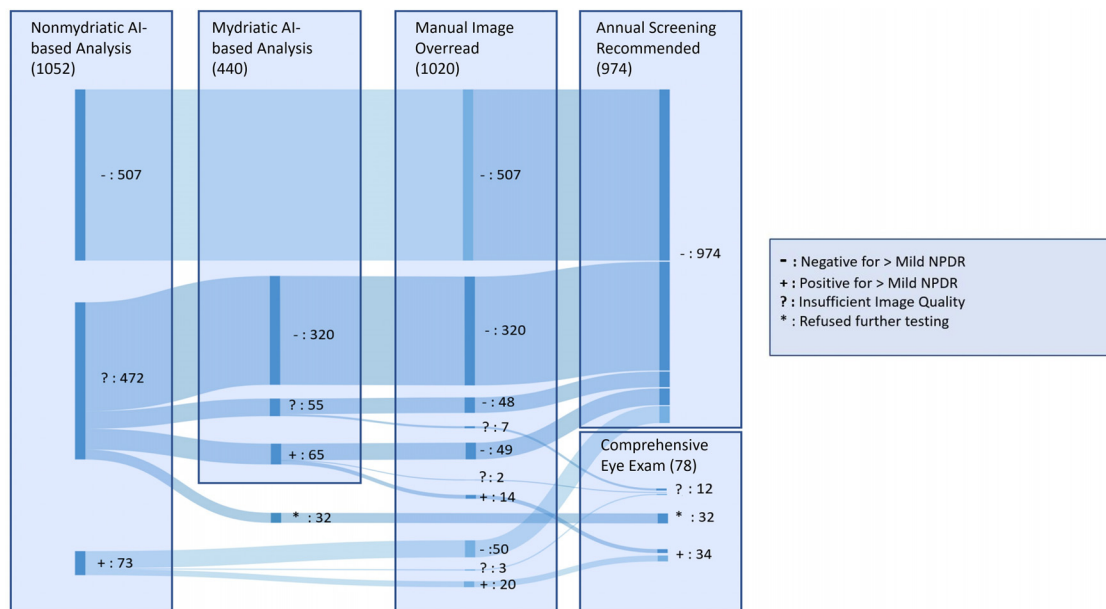


FIGURE 1. Flow diagram of diabetic retinopathy (DR) telemedicine screening results and recommendations. Results from each stage of DR screening including nonmydriatic artificial intelligence (AI)-based analysis, mydriatic AI-based analysis (when required), manual image overread, and recommendation for either continued annual telemedicine screening or more urgent comprehensive eye exam. At each stage, “-” indicates mild or absent non-proliferative diabetic retinopathy, “+” indicates greater than mild non-proliferative diabetic retinopathy, “?” indicates image quality insufficient for analysis, and “*” indicates patient declined further testing.

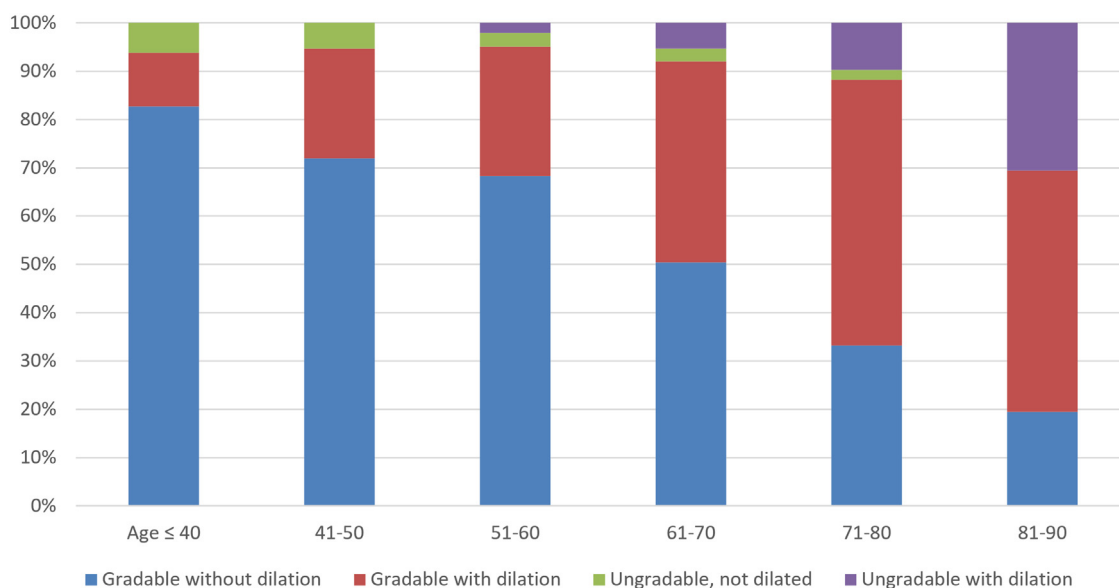


FIGURE 2. Proportion of patients with gradable mydriatic and nonmydriatic images by age. “Gradable without dilation” indicates that nonmydriatic fundus photos were of sufficient quality for artificial intelligence (AI)-based analysis. “Gradable with dilation” indicates that nonmydriatic fundus photos were not gradable, but mydriatic fundus photos were of sufficient quality for AI-based grading. “Ungradable, not dilated” indicates that nonmydriatic fundus photos were not gradable and patient declined pupillary dilation. “Ungradable with dilation” indicates that both nonmydriatic and mydriatic fundus photos were of insufficient quality for AI-based analysis.

TABLE 2. Image Gradeability by Age, Presence of Media Opacity

	n (%)	Gradable (%) Nonmydriatic	P Value	Gradable (%) Dilated	P Value	Ungradable (%) Declined Dilation
Overall	1052	580 (55)	—	965 (92)	—	32 (3)
Age, years						
<40	81 (8)	67 (83)	<i>ref</i>	76 (94)	<i>ref</i>	5 (6)
41-50	114 (11)	82 (72)	.08	108 (95)	.78	6 (5)
51-60	246 (24)	168 (68)	.01	234 (95)	.65	7 (3)
61-70	379 (37)	191 (50)	< .001	349 (92)	.59	10 (3)
71-80	196 (19)	65 (33)	< .001	173 (88)	.16	4 (2)
>80	36 (3)	7 (19)	< .001	25 (69)	< .001	0
Media opacity?						
None	387 (36)	234 (60)	<i>ref</i>	361 (93)	<i>ref</i>	15 (4)
Yes, Lens/IOL	221 (21)	80 (36)	< .001	185 (84)	< .001	5 (2)
Yes, Other	49 (5)	21 (43)	< .001	43 (88)	< .001	2 (4)
—	390 (37)					

Dilated = patients with gradable images after pupillary dilation protocol; lens/IOL = media opacity involving crystalline lens or implanted intraocular lens; nonmydriatic = patients with gradable images prior to pupillary dilation protocol; other media opacity not involving crystalline lens or implanted intraocular lens.

80 years (33.2%; $P < .001$), and 7 of 36 patients aged 81 to 90 years (19.4%; $P < .001$) (Table 2).

Screening photographs for patients without previously documented ocular media opacity were gradable in 361 of 387 patients (93.2%) by the AI-based platform compared with 185 of 221 (83.7%; $P < .001$) of those with previously documented lens/IOL opacities, which were defined as a minimum of 2+ nuclear lens opacity, 2+ cortical changes, 1+ posterior capsular cataract, or 1+ posterior capsular opacification. Forty-three of 49 patients (87.7%) patients with “other” media opacity (eg, corneal scarring, asteroid hyalosis) had gradable fundus photographs ($P < .001$ vs no opacity) (Table 2). Patients without documented media opacity were also more likely to have gradable nonmydriatic imaging (60.5%, 234 of 387) compared with those with lens/IOL opacities (36.2%, 80 of 221; $P < .001$) or “other” media opacities (42.8%, 21 of 49, $P < .001$).

Analyses of the potential impact of demographic factors and diabetes type on image gradeability revealed relatively few differences (Table 1). Female patients were more likely to have gradable nonmydriatic images (60% vs 52%; $P = .02$) but overall image gradeability between sexes was similar ($P = .49$). Eight hundred fifteen patients (78%) receiving DR screening self-identified as white, which is consistent with overall demographics of southeast Minnesota. Patients who self-identified as Hispanic or Black had similar rates of IDx-DR gradable imaging (both overall and nonmydriatic), but those who self-identified as “other” were less likely to have gradable imaging overall (84% vs 93%; $P < .001$). Patients with type 1 diabetes were more likely to have gradable nonmydriatic imaging compared with those with type 2 diabetes (76.4% vs 53.6%; $P = .001$); however, there was no significant difference after adjustment for age and no difference was noted following protocol dilation.

DISCUSSION

Continuous improvement in the quality and utilization of DR screening systems is an important public health priority. Sight-threatening retinopathy often develops prior to any visual symptoms, and long-term vision outcomes are more favorable with early management when retinopathy progresses beyond an evidence-based threshold for treatment. Systemic health management, patient education, and communication are critically important for reducing diabetes-associated ocular complications, but DR screening systems must also continue to lower barriers for patients to receive regular high-quality screening.

Co-location of retinal screening with other medical appointments, as well as limitation of pharmacologic pupillary dilation to only those who require it, may help increase adherence to recommended DR screening regimens. The current study reports real-world outcomes following the introduction of AI-based image analysis into an existing DR telemedicine system. Immediate point-of-care screening results offered by the AI-based platform enabled the establishment of a reflex dilation protocol, and 55% of patients were successfully screened without dilation. Sufficient image quality was maintained with reflex dilation to effectively screen 92% of all patients who presented for DR screening within the primary care setting.

The introduction of AI-based image analysis into this established DR telemedicine program resulted in excellent sensitivity (100%; 95% CI 90.8-100.00%) for the detection of greater than mild DR requiring further evaluation, which aligns with previously reported sensitivity results for the IDx-DR system (range 87-98%).¹⁹⁻²³ Screening systems have several options for managing positive results, including immediate referral of all patients for com-

prehensive exam within a relatively short timeline, or to have a secondary review, as in the current system, to triage both follow-up timing and setting (eg, general eye exam vs retina specialist) based on the assessed level of risk. If a DR telemedicine system goal is to minimize the number of patient images reviewed by a trained analyst, automated interpretation must have a very low rate of false negative results. Of the 827 patients whose fundus imaging was graded as “negative” in this study, there was 100% agreement on manual overread. The specificity was somewhat lower at 89.2%, though not dissimilar to previously reports ranging from 84-93.8%.¹⁹⁻²³

The 27.5% positive predictive value of IDx-DR graded images in this study was lower than the 73% PPV reported by Abramoff and associates in 2018,²² but similar to the 30% PPV reported by van der Heijden and associates, and 39.7% PPV reported by Verbraak and associates.²¹⁻²³ These differences are likely multifactorial and related to differences in reference grading (clinical assessment for referable retinopathy vs fundus reading center grading), as well as to differences in screening populations and telemedicine program structures. It is possible that clinical assessments of DR in the current study had a higher threshold for referable retinopathy compared with the greater than mild DR threshold applied by the clinical trial’s fundus reading center. Another potential factor is that PPV may be lower in study populations where a higher proportion of patients graded positive by the IDx-DR system have DR levels relatively close to the greater than mild DR threshold. Manual grading is more likely to agree with positive AI-based interpretation results when DR is more advanced. Of note, the clinical trial population in which PPV was 73% was enriched with patients at higher risk of more severe retinopathy; 23.8% of trial subjects had referable DR compared with 3.2% of patients in the current study.²² This difference in study populations and other potential differences in group demographics may have impacted the results. Differences in the structure of screening programs, such as the recommendation for comprehensive dilated eye exams every 3 years in the current telemedicine program, may have also altered screening populations and therefore results.

Although the PPV reported in this study was relatively low, integration of AI-based image interpretation into the existing telemedicine system significantly reduced the number of patient images requiring prompt manual review. Furthermore, the very high sensitivity and negative predictive value of initial AI-based grading within the system potentially supports restricting secondary review to only positive AI-based image interpretations. Over this study period, a protocol employing clinician review of only ungradable images and initial positive screens (225/1052) would have reduced clinician image interpretation by 79%.

In this study, gradeability of both nonmydriatic and mydriatic images was highly correlated with patient age, with significantly higher proportions of patients in the older

age groups both requiring reflex dilation and continuing to have insufficient image quality following protocol dilation (Figure 2). It is well-established that pupil size decreases with age, likely related to decreased sympathetic tone.^{24,25} Previous studies have similarly identified increasing age as an important factor in DR screening fundus photograph quality, with or without dilation.²⁵⁻²⁷ In these studies, the presence of ocular media opacity, primarily cataract, was associated with a lower likelihood of sufficient image quality for screening.²⁵⁻²⁷ The current results corroborate those findings, as a significantly lower proportion of patients with previously documented ocular media opacity had successful AI-based image analysis.

Limitations of this study included the relatively homogeneous study population and higher proportion of Caucasian patients compared with the overall global population of individuals with diabetes. These data may have also, due to the nature of recruitment in the primary care setting, over-represented patients with good adherence to recommended primary care follow-up. These factors may have limited generalizability to other populations and settings. Other potential limitations are related to the secondary overread process. All image interpretation was performed in accordance with established local telemedicine practice patterns and referral thresholds; however, a formal fundus photograph reading center would have offered more precise secondary DR severity scale assessments. Secondary overread of IDx-DR graded images was also performed by reviewers who were not masked to the original grading, which may have introduced bias. Furthermore, positive and negative graded images were reviewed in different settings, with positive images being promptly reviewed in relatively small numbers and negative images being non-urgently batch-reviewed in larger numbers. Finally, some features of the Mayo Clinic Diabetic Retinopathy Telemedicine Program, such as the standard recommendation for comprehensive eye examinations every third year, may have also limited generalizability to systems with different established practice patterns.

AI-based autonomous image analysis is transitioning from “promise” to “reality” across many realms. Within medicine, ophthalmology and retinal diseases are at the forefront to benefit from advances in AI-based diagnostic capabilities, and the IDx-DR system is the first such FDA-approved AI-based autonomous diagnostic platform to enter routine clinical practice. Following integration of AI-based image analysis into an existing DR telemedicine program, this study reports real-world results demonstrating high sensitivity for detection of greater than mild DR, as well as the benefits offered by real-time results including optimizing pharmacologic pupillary dilation for only those who require it. Due to the significant drop-off in nonmydriatic image gradeability with increasing age, some screening systems may elect to pre-emptively dilate all patients above a specific age to improve efficiency; however, select patients of all ages are able to benefit from nonmydriatic DR screening.

REFERENCES

1. Teo ZL, Tham YC, Yu M, et al. Global prevalence of diabetic retinopathy and projection of burden through 2045: Systematic review and meta-analysis. *Ophthalmology*. 2021;128(11):1580–1591. doi:[10.1016/j.ophtha.2021.04](https://doi.org/10.1016/j.ophtha.2021.04).
2. GBD 2019 Blindness and Vision Impairment Collaborators; Vision Loss Expert Group of the Global Burden of Disease Study. Causes of blindness and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to VISION 2020: The Right to Sight: An analysis for the Global Burden of Disease Study [published correction appears in *Lancet Glob Health*. 2021;9(4):e408]. *Lancet Glob Health*. 2021;9(2):e144–e160. doi:[10.1016/S2214-109X\(20\)30489-7](https://doi.org/10.1016/S2214-109X(20)30489-7).
3. Vujosevic S, Aldington SJ, Silva P, et al. Screening for diabetic retinopathy: New perspectives and challenges. *Lancet Diabetes Endocrinol*. 2020;8(4):337–347. doi:[10.1016/S2213-8587\(19\)30411-5](https://doi.org/10.1016/S2213-8587(19)30411-5).
4. Cheung N, Mitchell P, Wong TY. Diabetic retinopathy. *Lancet*. 2010;376(9735):124–136. doi:[10.1016/S0140-6736\(09\)62124-3](https://doi.org/10.1016/S0140-6736(09)62124-3).
5. Zhang X, Saaddine JB, Chou CF, et al. Prevalence of diabetic retinopathy in the United States, 2005–2008. *JAMA*. 2010;304(6):649–656. doi:[10.1001/jama.2010.1111](https://doi.org/10.1001/jama.2010.1111).
6. Hammes HP, Kerner W, Hofer S, et al. Diabetic retinopathy in type 1 diabetes—a contemporary analysis of 8,784 patients. *Diabetologia*. 2011;54(8):1977–1984. doi:[10.1007/s00125-011-2198-1](https://doi.org/10.1007/s00125-011-2198-1).
7. Roy MS, Klein R, O'Colmain BJ, Klein BEK, Moss SE, Kempen JH. The prevalence of diabetic retinopathy among adult type 1 diabetic persons in the United States. *Arch Ophthalmol*. 2004;122(4):546–551. doi:[10.1001/archophth.122.4.546](https://doi.org/10.1001/archophth.122.4.546).
8. Sabanayagam C, Banu R, Chee ML, et al. Incidence and progression of diabetic retinopathy: a systematic review. *Lancet Diabetes Endocrinol*. 2019;7(2):140–149. doi:[10.1016/S2213-8587\(18\)30128-1](https://doi.org/10.1016/S2213-8587(18)30128-1).
9. Yau JWY, Rogers SL, Kawasaki R, et al. Global prevalence and major risk factors of diabetic retinopathy. *Diabetes Care*. 2012;35(3):556–564. doi:[10.2337/dc11-1909](https://doi.org/10.2337/dc11-1909).
10. Leasher JL, Bourne RRA, Flaxman SR, et al. Global estimates on the number of people blind or visually impaired by diabetic retinopathy: A meta-analysis from 1990 to 2010. *Diabetes Care*. 2016;39(9):1643–1649. doi:[10.2337/dc15-2171](https://doi.org/10.2337/dc15-2171).
11. Barkmeier AJ. Toward optimal screening for diabetic retinopathy: Balancing precision and pragmatism. *Mayo Clin Proc*. 2021;96(2):282–284. doi:[10.1016/j.mayocp.2020.12.008](https://doi.org/10.1016/j.mayocp.2020.12.008).
12. Wong TY, Sun J, Kawasaki R, et al. Guidelines on diabetic eye care. *Ophthalmology*. 2018;125(10):1608–1622. doi:[10.1016/j.ophtha.2018.04.007](https://doi.org/10.1016/j.ophtha.2018.04.007).
13. Ferris FL. How effective are treatments for diabetic retinopathy? *JAMA*. 1993;269(10):1290. doi:[10.1001/jama.1993.03500100088034](https://doi.org/10.1001/jama.1993.03500100088034).
14. Flaxel CJ, Adelman RA, Bailey ST, et al. Diabetic retinopathy preferred practice pattern®. *Ophthalmology*. 2020;127(1):P66–P145. doi:[10.1016/j.ophtha.2019.09.025](https://doi.org/10.1016/j.ophtha.2019.09.025).
15. Solomon SD, Chew E, Duh EJ, et al. Diabetic Retinopathy: A position statement by the American Diabetes Association. *Diabetes Care*. 2017;40(3):412–418. doi:[10.2337/dc16-2641](https://doi.org/10.2337/dc16-2641).
16. Kashim RM, Newton P, Ojo O. Diabetic retinopathy screening: A systematic review on patients' non-attendance. *Int J Environ Res Public Health*. 2018;15(1):157. doi:[10.3390/ijerph15010157](https://doi.org/10.3390/ijerph15010157).
17. Ting DSW, Cheung GCM, Wong TY. Diabetic retinopathy: global prevalence, major risk factors, screening practices and public health challenges: a review: Global burden of diabetic eye diseases. *Clin Experiment Ophthalmol*. 2016;44(4):260–277. doi:[10.1111/ceo.12696](https://doi.org/10.1111/ceo.12696).
18. Gunasekaran DV, Ting DSW, Tan GSW, Wong TY. Artificial intelligence for diabetic retinopathy screening, prediction and management. *Curr Opin Ophthalmol*. 2020;31(5):357–365. doi:[10.1097/ICU.0000000000000693](https://doi.org/10.1097/ICU.0000000000000693).
19. Grzybowski A, Brona P, Lim G, et al. Artificial intelligence for diabetic retinopathy screening: a review. *Eye*. 2020;34(3):451–460. doi:[10.1038/s41433-019-0566-0](https://doi.org/10.1038/s41433-019-0566-0).
20. Abramoff MD, Lou Y, Erginay A, et al. Improved automated detection of diabetic retinopathy on a publicly available dataset through integration of deep learning. *Invest Ophthalmol Vis Sci*. 2016;57(13):5200–5206. doi:[10.1167/iovs.16-19964](https://doi.org/10.1167/iovs.16-19964).
21. der Heijden AA van, MD Abramoff, Verbraak F, van Hecke MV, Liem A, Nijpels G. Validation of automated screening for referable diabetic retinopathy with the IDx-DR device in the Hoorn Diabetes Care System. *Acta Ophthalmol (Copenh)*. 2018;96(1):63–68. doi:[10.1111/aos.13613](https://doi.org/10.1111/aos.13613).
22. Abramoff MD, Lavin PT, Birch M, Shah N, Folk JC. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *NPJ Digit Med*. 2018;1(1):1–8. doi:[10.1038/s41746-018-0040-6](https://doi.org/10.1038/s41746-018-0040-6).
23. Verbraak FD, Abramoff MD, Bausch GCF, et al. Diagnostic accuracy of a device for the automated detection of diabetic retinopathy in a primary care setting. *Diabetes Care*. 2019;42(4):651–656. doi:[10.2337/dc18-0148](https://doi.org/10.2337/dc18-0148).
24. Korczyn AD, Laor N, Nemet P. Sympathetic pupillary tone in old age. *Arch Ophthalmol*. 1976;94(11):1905–1906. doi:[10.1001/archophth.1976.03910040615006](https://doi.org/10.1001/archophth.1976.03910040615006).
25. Scanlon PH, Foy C, Malhotra R, Aldington SJ. The influence of age, duration of diabetes, cataract, and pupil size on image quality in digital photographic retinal screening. *Diabetes Care*. 2005;28(10):2448–2453. doi:[10.2337/diacare.28.10.2448](https://doi.org/10.2337/diacare.28.10.2448).

26. Murgatroyd H. Effect of mydriasis and different field strategies on digital image screening of diabetic eye disease. *Br J Ophthalmol*. 2004;88(7):920–924. doi:[10.1136/bjo.2003.026385](https://doi.org/10.1136/bjo.2003.026385).
27. Penman AD, Saaddine JB, Hegazy M, et al. Screening for diabetic retinopathy: the utility of nonmydriatic retinal photography in Egyptian adults. *Diabet Med J Br Diabet Assoc*. 1998;15(9):783–787. The PMID is 9737809. doi:[10.1002/\(SICI\)1096-9136\(199809\)15:9<783::AID-DIA634>3.0.CO;2-5](https://doi.org/10.1002/(SICI)1096-9136(199809)15:9<783::AID-DIA634>3.0.CO;2-5).